EE2015: Electric Circuits and Networks

Tutorial 7 Answer Key

(October 3, 2025)

1. (a) $v_C(t) = -0.2083\,e^{-2t} + 5.208\,e^{-50t}$
   (b) $v_C(t) = (5 - 50t)\,e^{-10t}$
   (c) $v_C(t) = (5\cos 6t - 6.6667\sin 6t)\,e^{-8t}$
2. $R = 750\,\Omega$, $L = 25\,\text{H}$, $C = 200\,\mu\text{F}$
3. (a) $i_L(0^+) = 0$, $v_C(0^+) = -10\,\text{V}$, $v_R(0^+) = 0\,\text{V}$
   (b) $i'_L(0^+) = 0\,\text{A/s}$, $v'_C(0^+) = 8\,\text{V/s}$, $v'_R(0^+) = 8\,\text{V/s}$
   (c) $i_L(\infty) = 0.4\,\text{A}$, $v_C(\infty) = 6\,\text{V}$, $v_R(\infty) = 16\,\text{V}$
4. $R = 120\,\Omega$
5. $v_C(t) = 18e^{-t} - 2e^{-9t}$ for $t \geq 0$
6. $v_C(t) = 50 + e^{-3t}(-62\cos 4t - 46.5\sin 4t)$ for $t \geq 0$
7. $i_L(t) = 7.5\,e^{-4t}\,u(t)\,\text{A}$
8. $v(t) = 20 + 47.46\,e^{-4.79t} + 12.54\,e^{-0.21t}\,\text{V}$